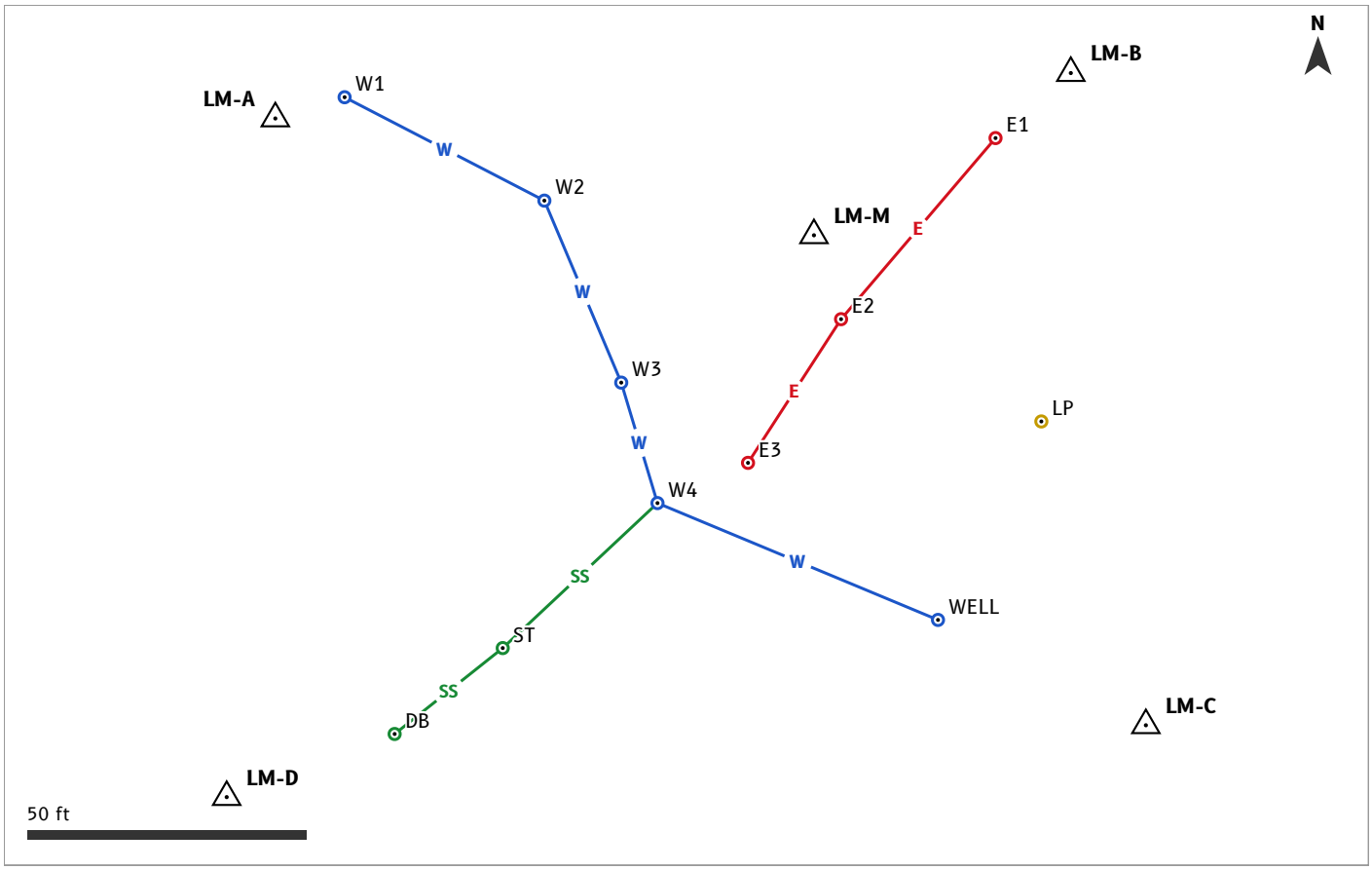
Example property

1234 Example Street

Surveyed by D. Panzino · Report printed 2026-10-15



Datum and solution

Coordinates	Local, in feet. LM-A held at N 5000.00, E 5000.00.
Basis of bearings	LM-A to LM-B = N 86°44'09" E (grid north).
Measurements	48 angles, 2 distances. Reading precision assumed 20" (one pointing); distances ±(0.02 ft + 1:5000).
Solution	16 of 16 located. 21 redundant measurements; readings fit at 0.27× the assumed precision.

Landmarks (control)

Name	Northing	Easting	±95%	Description / how to find it
LM-A	5000.00	5000.00	0.00	1/2" rebar with yellow cap, 2 ft E of driveway apron, NW corner
LM-B	5008.12	5142.36	0.08	1/2" rebar with yellow cap at NE fence corner post
LM-C	4891.46	5155.79	0.11	Chiseled X on concrete pad of shed, SE corner
LM-D	4878.61	4991.30	0.07	1/2" rebar with yellow cap, SW corner by garden
LM-M	4979.10	5096.39	0.06	16d nail in top of deck post, NW post of rear deck

Points

Name	Type	Northing	Easting	±95%	Check	Depth	Description
W1	Water	5003.80	5012.40	0.04	yes	42 in	Water meter box at street
W2	Water	4985.30	5048.15	0.05	yes	42 in	Water line bend (45 deg)
W3	Water	4952.75	5061.91	0.05	no	36 in	Water line tee to hose bib

W4	Water	4931.20	5068.39	0.06	yes	36 in	Water entry at house foundation
E1	Electric	4996.49	5128.89	0.08	yes	30 in	Transformer pedestal
E2	Electric	4964.10	5101.24	0.06	yes	24 in	Conduit bend
E3	Electric	4938.40	5084.59	0.06	yes	24 in	House meter base, conduit up
ST	Sewer / septic	4905.27	5040.70	0.07	yes	18 in to lid	Septic tank inlet lid
DB	Sewer / septic	4889.90	5021.35	0.08	yes	20 in	Septic distribution box
WELL	Water	4910.30	5118.59	0.09	yes		Well head (casing)
LP	Gas / oil	4945.80	5137.09	0.09	yes		Propane tank fill valve

"Check" = the point is seen from enough different directions that a misread angle would have shown up as a flagged reading.

Lines

Name	Type	Route	Length	Depth	Description
Water service	Water	W1 → W2 → W3 → W4	98.1 ft	36-42 in	1" PE
Electric service	Electric	E1 → E2 → E3	73.2 ft	24-30 in	2" PVC, 200A
Septic	Sewer / septic	W4 → ST → DB	62.6 ft		4" PVC
Well line	Water	WELL → W4	54.4 ft	48 in	1" PE

Field book

Date	Station	Backsight	Target	Reading	Angle right	Distance	Residual	Note
2026-10-10	**LM-A**	LM-B @ 0 00 00	LM-C	38 07 40	38°07'40"		-5"	
			LM-D	97 22 00	97°22'00"		3"	
			LM-M	15 29 40	15°29'40"		-10"	
			W1	346 13 40	346°13'40"		0"	
			W3	40 37 00	40°37'00"		-1"	
			W4	48 26 00	48°26'00"		-3"	
			E1	4 49 20	4°49'20"		0"	
			E2	22 47 20	22°47'20"		0"	
			E3	39 19 40	39°19'40"		2"	
			ST	70 01 00	70°01'00"		3"	
			DB	82 17 20	82°17'20"		1"	
			WELL	40 22 00	40°22'00"		3"	
2026-10-11	**LM-B**	LM-A @ 0 00 00	LM-C	266 41 40	266°41'40"		-5"	
			LM-D	322 39 40	322°39'40"		12"	
			LM-M	331 00 00	331°00'00"		-9"	
			W2	349 39 00	349°39'00"		-1"	
			W3	328 43 40	328°43'40"		-5"	
			E1	322 28 40	322°28'40"		0"	
			E2	316 18 40	316°18'40"		0"	
			WELL	286 55 20	286°55'20"		-2"	
			LP	278 06 00	278°06'00"		2"	
2026-10-12	**LM-C**	LM-B @ 0 00 00	LM-A	311 26 00	311°26'00"		1"	
			LM-D	272 06 00	272°06'00"		-8"	
			LM-M	332 26 20	332°26'20"		-3"	
			W1	314 38 40	314°38'40"		0"	
			W2	317 39 00	317°39'00"		1"	
			E1	352 12 00	352°12'00"		0"	
			ST	283 24 40	283°24'40"		4"	
			DB	275 54 20	275°54'20"		2"	

2026-10-13	**LM-D**	LM-A @ 0 00 00	LM-B	45 17 40	45°17'40"		9"	
			LM-C	81 26 00	81°26'00"		4"	
			LM-M	42 10 40	42°10'40"		-14"	
			W1	5 28 00	5°28'00"		0"	
			W2	23 57 00	23°57'00"		-1"	
			W3	39 30 00	39°30'00"		-5"	
			W4	51 36 00	51°36'00"		7"	
			E3	53 14 40	53°14'40"		-3"	
			WELL	71 55 00	71°55'00"		-2"	
			LP	61 09 20	61°09'20"		4"	
2026-10-14	**LM-M**	LM-A @ 0 00 00	LM-B	135 30 20	135°30'20"		1"	
			LM-C	223 38 20	223°38'20"		1"	
			LM-D	304 03 00	304°03'00"		-1"	
			W4	288 04 40	288°04'40"		4"	
			E2	239 50 40	239°50'40"		0"	
			E3	273 56 00	273°56'00"		-1"	
			ST	294 47 40	294°47'40"		-3"	
			DB	297 50 20	297°50'20"		-2"	
			LP	207 03 20	207°03'20"		-1"	
	Taped: LM-A – LM-B					142.61	0.018 ft	Steel tape, 2 measurements averaged
	Taped: LM-C – LM-D					164.98	-0.019 ft	Steel tape

How to find each point

W1 — Water meter box at street N 5003.80 E 5012.40

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **346°13'40"**; the point is **12.97 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **5°28'00"**; the point is **126.95 ft** out along that line.

Where the two lines cross: expected within ±0.05 ft (95%). Tape check: 87.55 ft from LM-M and 126.95 ft from LM-D; 12.97 ft from LM-A.

W2 — Water line bend (45 deg) N 4985.30 E 5048.15

1. Set up on **LM-C**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **317°38'59"**; the point is **142.81 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **23°57'01"**; the point is **120.89 ft** out along that line.

Where the two lines cross: expected within ±0.05 ft (95%). Tape check: 50.34 ft from LM-A and 120.89 ft from LM-D; 48.64 ft from LM-M.

W3 — Water line tee to hose bib N 4952.75 E 5061.91

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **40°37'01"**; the point is **77.88 ft** out along that line.
2. Set up on **LM-D**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **39°30'05"**; the point is **102.38 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). Tape check: 43.40 ft from LM-M and 77.88 ft from LM-A; 97.67 ft from LM-B.

W4 — Water entry at house foundation N 4931.20 E 5068.39

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **48°26'03"**; the point is **97.01 ft** out along that line.
2. Set up on **LM-M**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **288°04'36"**; the point is **55.48 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). Tape check: 55.48 ft from LM-M and 96.01 ft from LM-C; 93.32 ft from LM-D.

E1 — Transformer pedestal N 4996.49 E 5128.89

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **322°28'40"**; the point is **17.80 ft** out along that line.
2. Set up on **LM-C**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **352°12'00"**; the point is **108.43 ft** out along that line.

Where the two lines cross: expected within ±0.04 ft (95%). Tape check: 36.86 ft from LM-M and 108.43 ft from LM-C; 17.80 ft from LM-B.

E2 — Conduit bend N 4964.10 E 5101.24

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **316°18'40"**; the point is **60.24 ft** out along that line.
2. Set up on **LM-M**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **239°50'40"**; the point is **15.77 ft** out along that line.

Where the two lines cross: expected within ±0.02 ft (95%). Tape check: 15.77 ft from LM-M and 60.24 ft from LM-B; 90.84 ft from LM-C.

E3 — House meter base, conduit up N 4938.40 E 5084.59

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **39°19'38"**; the point is **104.64 ft** out along that line.
2. Set up on **LM-M**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **273°56'01"**; the point is **42.38 ft** out along that line.

Where the two lines cross: expected within ±0.04 ft (95%). Tape check: 42.38 ft from LM-M and 85.28 ft from LM-C; 90.54 ft from LM-B.

ST — Septic tank inlet lid N 4905.27 E 5040.70

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **70°00'57"**; the point is **103.10 ft** out along that line.
2. Set up on **LM-M**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **294°47'43"**; the point is **92.48 ft** out along that line.

Where the two lines cross: expected within ±0.05 ft (95%). Tape check: 56.14 ft from LM-D and 103.10 ft from LM-A; 92.48 ft from LM-M.

DB — Septic distribution box N 4889.90 E 5021.35

1. Set up on **LM-A**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **82°17'19"**; the point is **112.15 ft** out along that line.
2. Set up on **LM-C**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **275°54'18"**; the point is **134.45 ft** out along that line.

Where the two lines cross: expected within ±0.05 ft (95%). Tape check: 32.11 ft from LM-D and 112.15 ft from LM-A; 116.56 ft from LM-M.

WELL — Well head (casing) N 4910.30 E 5118.59

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **286°55'22"**; the point is **100.66 ft** out along that line.
2. Set up on **LM-C**, sight **LM-B** and set the circle to 0°00'00". Turn clockwise to **303°26'01"**; the point is **41.70 ft** out along that line. *(line of sight not yet checked)*

Where the two lines cross: expected within ±0.03 ft (95%). Tape check: 41.70 ft from LM-C and 72.29 ft from LM-M; 100.66 ft from LM-B.

LP — Propane tank fill valve N 4945.80 E 5137.09

1. Set up on **LM-B**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **278°05'58"**; the point is **62.54 ft** out along that line.
2. Set up on **LM-M**, sight **LM-A** and set the circle to 0°00'00". Turn clockwise to **207°03'21"**; the point is **52.58 ft** out along that line.

Where the two lines cross: expected within ±0.03 ft (95%). Tape check: 52.58 ft from LM-M and 62.54 ft from LM-B; 57.47 ft from LM-C.

Notes

1. Coordinates are local (assumed). Bearings are referenced to the line LM-A to LM-B.
2. Depths are to top of pipe/conduit, measured when the trench was open.